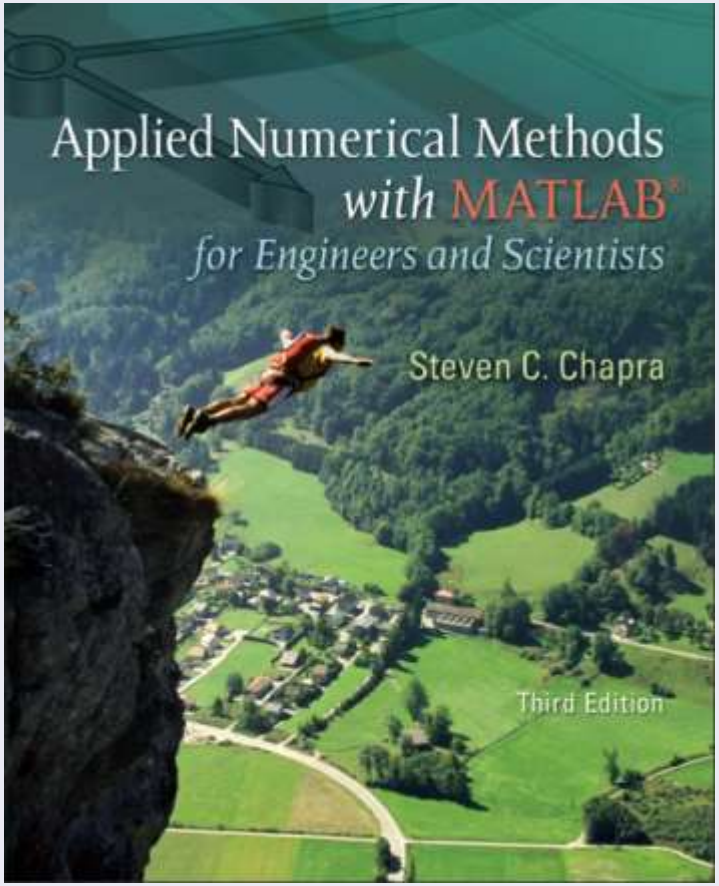


Part 4
Chapter 15

General Linear Least-Squares and Nonlinear Regression

PowerPoints organized by Dr. Michael R. Gustafson II, Duke University

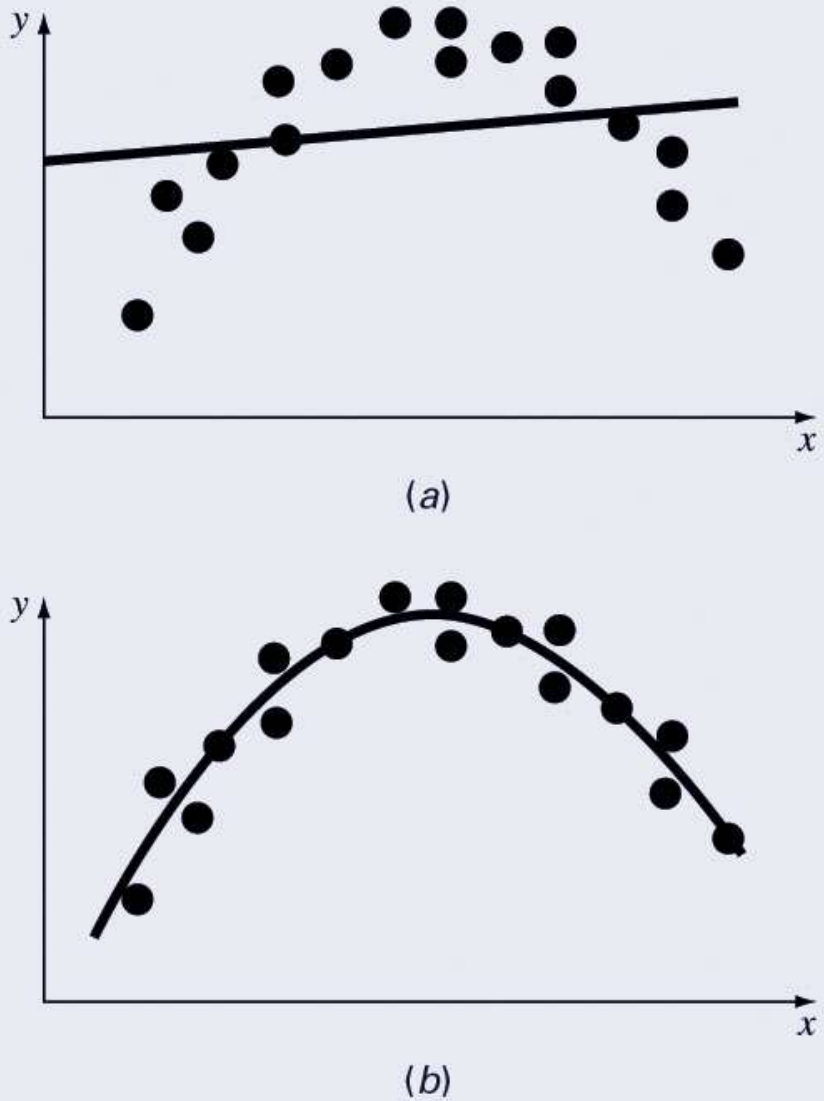
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Chapter Objectives

- Knowing how to implement polynomial regression.
- Knowing how to implement multiple linear regression.
- Understanding the formulation of the general linear least-squares model.
- Understanding how the general linear least-squares model can be solved with MATLAB using either the normal equations or left division.
- Understanding how to implement nonlinear regression with optimization techniques.

Polynomial Regression

- The least-squares procedure from Chapter 13 can be readily extended to fit data to a higher-order polynomial. Again, the idea is to minimize the sum of the squares of the estimate residuals.
- The figure shows the same data fit with:
  a) A first order polynomial
  b) A second order polynomial



Process and Measures of Fit

- For a second order polynomial, the best fit would mean minimizing:

$$S_r = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - a_0 - a_1 x_i - a_2 x_i^2 \right)^2$$

- In general, this would mean minimizing:

$$S_r = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - a_0 - a_1 x_i - a_2 x_i^2 - \cdots - a_m x_i^m \right)^2$$

- The standard error for fitting an mth order polynomial to n data points is:

$$s_{y/x} = \sqrt{\frac{S_r}{n - (m+1)}}$$

because the mth order polynomial has (m+1) coefficients.

- The coefficient of determination r^2 is still found using:

$$r^2 = \frac{S_t - S_r}{S_t}$$

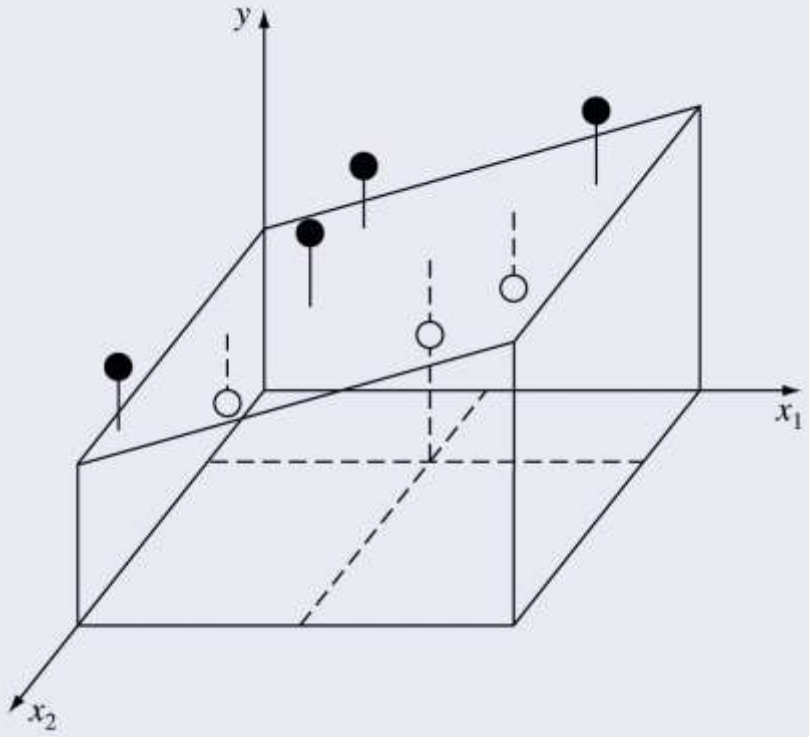
Multiple Linear Regression

- Another useful extension of linear regression is the case where *y* is a linear function of two or more independent variables:

$$y = a_0 + a_1 x_1 + a_2 x_2 + \cdots a_m x_m$$

- Again, the best fit is obtained by minimizing the sum of the squares of the estimate residuals:

$$S_r = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - a_0 - a_1 x_{1,i} - a_2 x_{2,i} - \cdots a_m x_{m,i} \right)^2$$



General Linear Least Squares

- Linear, polynomial, and multiple linear regression all belong to the general linear least-squares model:

$$y = a_0 z_0 + a_1 z_1 + a_2 z_2 + \cdots a_m z_m + e$$

  where z_0, z_1, …, z_m are a set of m+1 *basis functions* and e is the error of the fit.
- The basis functions can be any function data but *cannot* contain any of the coefficients a_0, a_1, etc.

Solving General Linear Least Squares Coefficients

- The equation:

$$y = a_0 z_0 + a_1 z_1 + a_2 z_2 + \cdots a_m z_m + e$$

can be re-written for each data point as a matrix equation:

$$\{y\} = [Z]\{a\} + \{e\}$$

where {*y*} contains the dependent data, {*a*} contains the coefficients of the equation, {*e*} contains the error at each point, and [*Z*] is:

$$[Z] = \begin{bmatrix} z_{01} & z_{11} & \cdots & z_{m1} \\ z_{02} & z_{12} & \cdots & z_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ z_{0n} & z_{1n} & \cdots & z_{mn} \end{bmatrix}$$

with z_{ji} representing the the value of the j_{th} basis function calculated at the i_{th} point.

Solving General Linear Least Squares Coefficients

- Generally, [*Z*] is not a square matrix, so simple inversion cannot be used to solve for {*a*}. Instead the sum of the squares of the estimate residuals is minimized:

$$S_r = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - \sum_{j=0}^{m} a_j z_{ji} \right)^2$$

- The outcome of this minimization yields:

$$\left[[Z]^T [Z] \right] \{a\} = \left\{ [Z]^T \{y\} \right\}$$

MATLAB Example

- Given *x* and *y* data in columns, solve for the coefficients of the best fit line for $y=a_0+a_1x+a_2x^2$

```
Z = [ones(size(x)) x x.^2]
a = (Z'*Z)\(Z'*y)
```

  - Note also that MATLAB's left-divide will automatically include the $[Z]^T$ terms if the matrix is not square, so

```
a = Z\y
```

    would work as well

- To calculate measures of fit:

```
St = sum((y-mean(y)).^2)
Sr = sum((y-Z*a).^2)
r2 = 1-Sr/St
syx = sqrt(Sr/(length(x)-length(a)))
```

Nonlinear Regression

- As seen in the previous chapter, not all fits are linear equations of coefficients and basis functions.
- One method to handle this is to transform the variables and solve for the best fit of the transformed variables. There are two problems with this method:
  - Not all equations can be transformed easily or at all
  - The best fit line represents the best fit for the transformed variables, not the original variables.
- Another method is to perform nonlinear regression to directly determine the least-squares fit.

Nonlinear Regression in MATLAB

- To perform nonlinear regression in MATLAB, write a function that returns the sum of the squares of the estimate residuals for a fit and then use MATLAB's `fminsearch` function to find the values of the coefficients where a minimum occurs.
- The arguments to the function to compute S_r should be the coefficients, the independent variables, and the dependent variables.

Nonlinear Regression in MATLAB

- Given dependent force data F for independent velocity data v, determine the coefficients for the fit:

$$F = a_0 v^{a_1}$$

$$S_r = \sum (y_i - f(x_i, \mathbf{a}))^2$$

- First - write a function called `fSSR.m` containing the following:

```
function f = fSSR(a, xm, ym)
yp = a(1)*xm.^a(2);
f = sum((ym-yp).^2);
```

- Then, use fminsearch in the command window to obtain the values of a that minimize `fSSR`:

```
a = fminsearch(@fSSR, [1, 1], [], v, F)
```

where [1, 1] is an initial guess for the [a0, a1] vector, [] is a placeholder for the options

Nonlinear Regression Results

- The resulting coefficients will produce the largest r^2 for the data and may be different from the coefficients produced by a transformation:

